Federated Learning for Brain AVM Segmentation
A real-world feasibility study
across 4 hospitals and 2 continents

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Multicenter 1 Year Pilot Study

Distributed, Collaborative, & Federated learning workshop

Research Collaboration Among Five Universities

- University of Illinois Chicago
- National Taiwan University, College of Medicine
- National Yang Ming Chiao Tung University, College of Medicine
- Taipei Medical University
- National Taiwan University of Science and Technology

4 Hospitals

UI health

National Taiwan University Hospital

Taipei Veterans General Hospital

Shuang Ho Hospital

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台大醫院 NTUH 國立台灣大學醫學院附設醫院 National Taiwan University Hospital

臺北榮民總醫院 Taipei Veterans General Hospital

衛生福利部雙和醫院 Taipei Medical University Shuang Ho Hospital

The Story Behind This Project



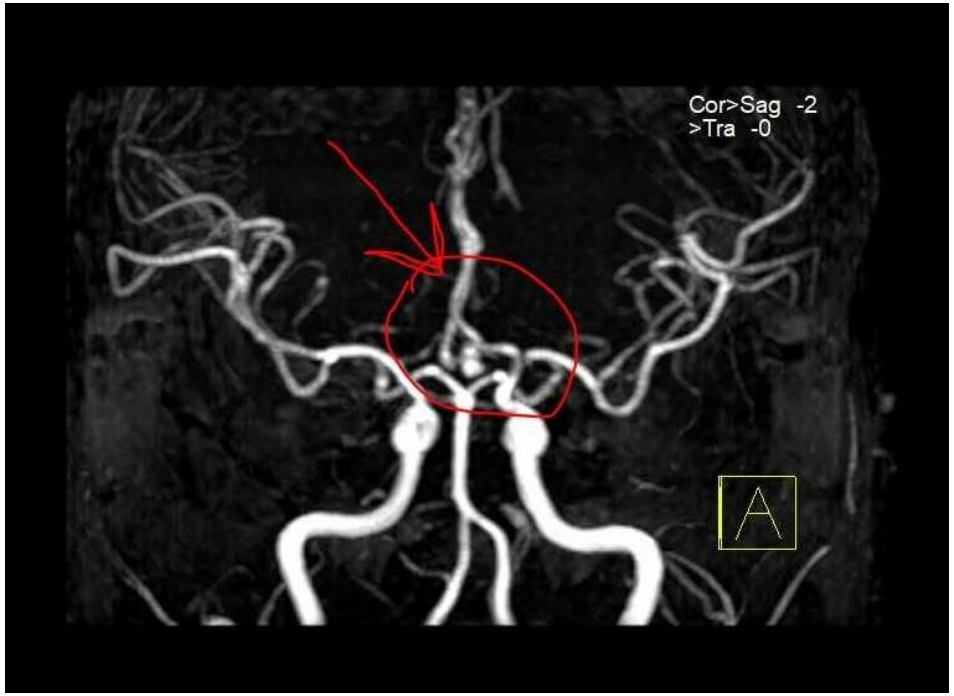
A Case of Missed Diagnosis in Mongolia
Brain aneurysm was initially diagnosed as hypertension



Mongolia

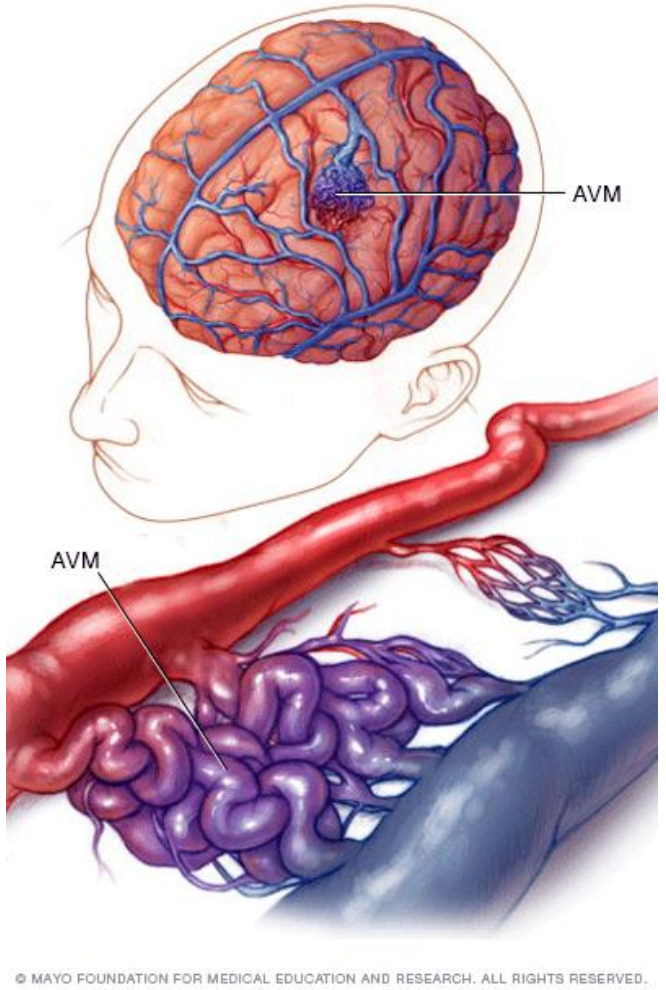


High misdiagnosis rates and limited access to medical specialists



Brain Arteriovenous Malformations (bAVM)

- Rare cerebrovascular lesions: 10–18 cases per 100,000 adults
- Abnormal artery-to-vein shunting through a complex vascular nidus
- Can present with intracranial hemorrhage, seizures, headache, or neurological deficits



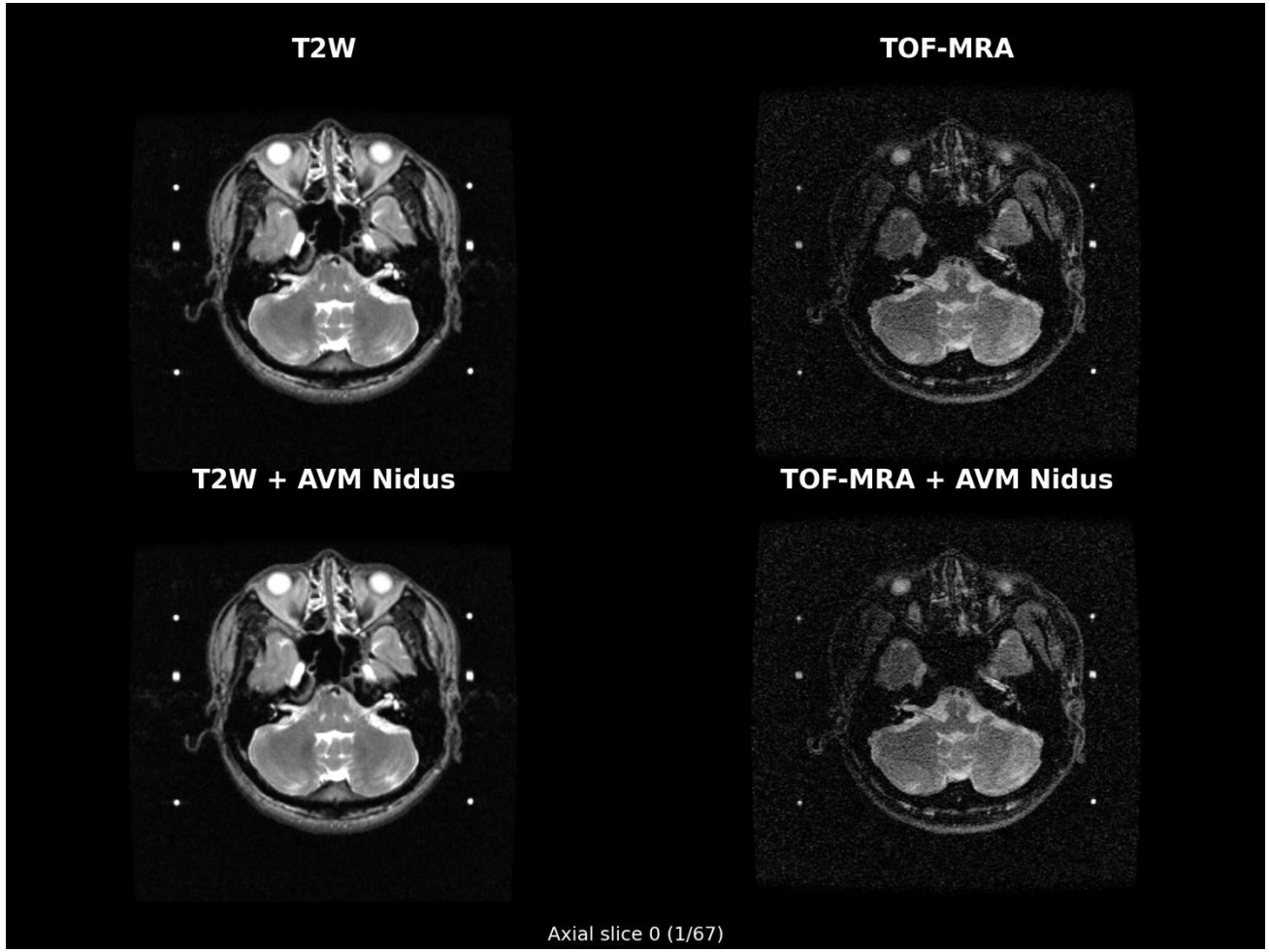
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Image source: https://www.beaconhealthsystem.org

Pan et al., JCBFM, 2021; Al-Shahi & Warlow, Brain, 2001

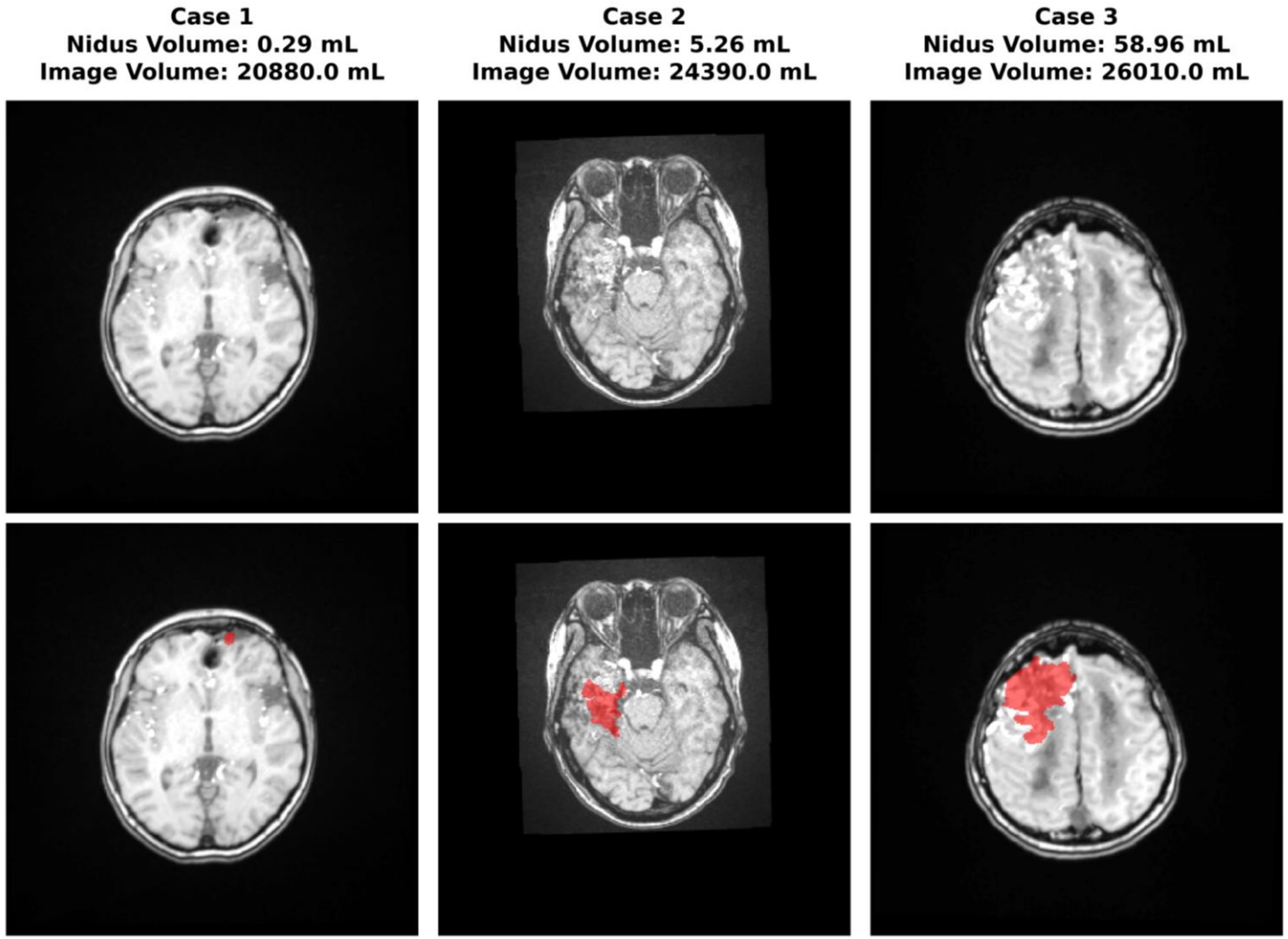
Accurate Segmentation is Important

- Treatment decisions depend on nidus size, location, venous drainage, and surrounding anatomy
- Accurate 3D delineation supports treatment planning, response assessment, and follow-up



Segmentation is Challenging

- Small, tortuous, fragmented, and heterogeneous nidus
- Ambiguous boundaries across MRI sequences
- Manual delineation is time-consuming and variable
- Imaging characteristics differ substantially across hospitals



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Why Multicenter Learning?

Distributed, Collaborative, & Federated learning workshop

1

Limited data

Each hospital sees relatively few bAVM cases, limiting coverage of lesion variability.

2

Domain shift

Scanner hardware, protocols, spatial resolution, and patient populations differ across sites.

3

Poor transfer

Strong performance at one hospital may not generalize to another.

4

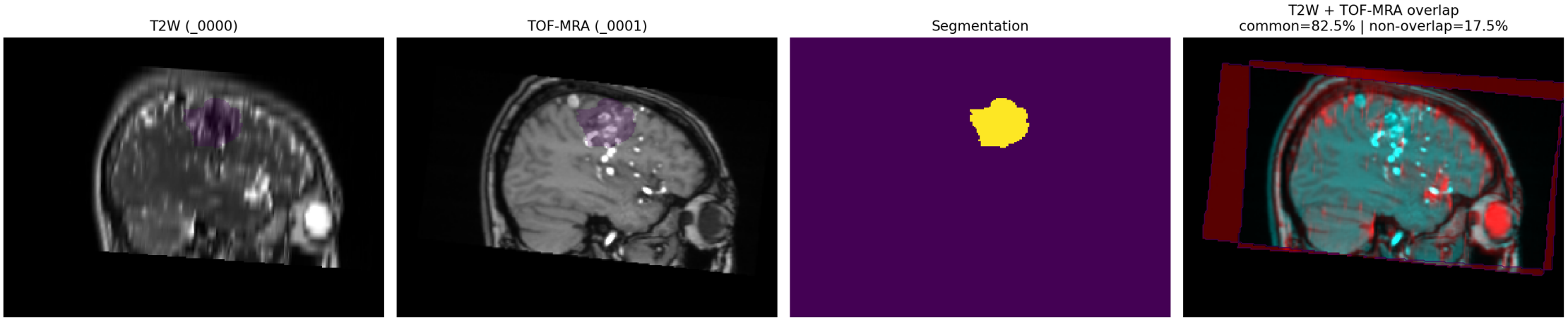
Governance constraints

Patient-level images and annotations cannot always be pooled centrally.

Federated learning addresses the data-sharing constraint by exchanging model updates while keeping patient-level data local.

Cohort and Study Design

- 200 patients from four hospitals
- 50 patients per hospital with paired T2W MRI and TOF-MRA
- 40 training + 10 held-out test cases at each site
- Segmentation target: treatment-planning bAVM nidus



AVMTS_010 | sagittal x-index=107 | RAS x=113.00 mm | largest mask area=1108 voxels | labels=0,1

MATERIALS & METHODS

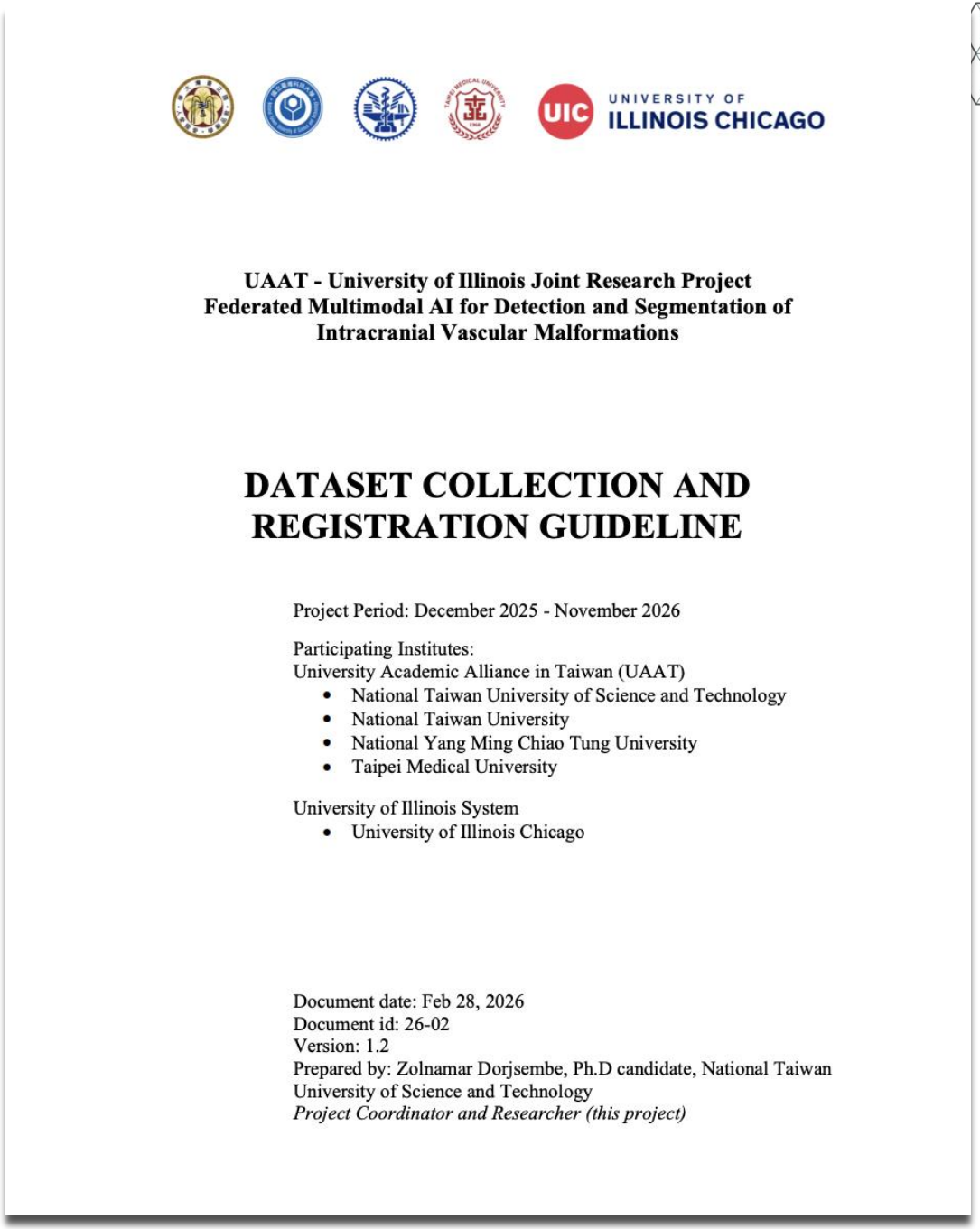
Patient and Imaging Characteristics by Institution

Distributed, Collaborative, & Federated learning workshop

Characteristic	H1	H2	H3	H4
Patient characteristics				
Patients, *n*	50	50	50	50
Age, years	36.3 [28.4–51.2]	47.0 [28.5–57.8]	43.0 [29.5–55.8]	40.0 [30.3–56.0]
Age range, years	20.0–63.2	10–71	6–89	14–74
Female	22 (44%)	23 (46%)	25 (50%)	24 (48%)
AVM laterality				
Left	30 (60%)	24 (48%)	27 (54%)	23 (46%)
Right	18 (36%)	21 (42%)	21 (42%)	25 (50%)
Midline or bilateral	0	5 (10%)	2 (4%)	2 (4%)
Unspecified	2 (4%)	0	0	0
AVM location				
Lobar	46 (92%)	33 (66%)	40 (80%)	37 (74%)
Deep	2 (4%)	6 (12%)	5 (10%)	8 (16%)
Infratentorial	2 (4%)	10 (20%)	4 (8%)	5 (10%)
Other or unspecified	0	1 (2%)	1 (2%)	0
Nidus volume, cm^3	13.5 [8.6–21.0]	6.9 [3.0–15.0]	6.2 [2.0–14.0]	4.4 [1.8–11.5]
Nidus-volume range, cm^3	1.4–44.6	0.3–47.1	0.3–59.0	0.2–68.8
Imaging characteristics				
Field strength	1.5 T	1.5 T	1.5 T	1.5 T and 3 T
MRI scanner models, *n*	1	2	14	6
TOF-MRA in-plane spacing, mm	0.500	0.508 [0.449–0.508]	0.586 [0.195–0.586]	0.430 [0.361–0.430]
TOF-MRA through-plane spacing, mm	1.500	1.600 [1.500–2.000]	0.600 [0.500–2.000]	1.400 [1.250–1.400]
T2-weighted in-plane spacing, mm	0.500	0.508	0.430 [0.286–0.937]	0.469
T2-weighted through-plane spacing, mm	1.500	2.000 [2.000–3.000]	5.000 [4.000–6.000]	1.400
Radiosurgery platform	Gamma Knife	Gamma Knife	CyberKnife	LINAC

Continuous variables are reported as median [interquartile range] (range), and categorical variables as n (%).

Data Collection Guideline



UAAT - University of Illinois Joint Research Project
Federated Multimodal AI for Detection and Segmentation of Intracranial Vascular Malformations

DATASET COLLECTION AND REGISTRATION GUIDELINE

Project Period: December 2025 - November 2026

Participating Institutes:
University Academic Alliance in Taiwan (UAAT)

- National Taiwan University of Science and Technology
- National Taiwan University
- National Yang Ming Chiao Tung University
- Taipei Medical University

University of Illinois System

- University of Illinois Chicago

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Prepared by: Zolnamar Dorjsembe, Ph.D candidate, National Taiwan University of Science and Technology
Project Coordinator and Researcher (this project)

Annotation Harmonization and Preprocessing

1	**Clinical target**	Treatment-planning bAVM nidus masks extracted from RTSS.
2	**Consensus**	Seven clinical experts agreed on common target definitions and boundary conventions.
3	**Quality review**	Biweekly cross-institution meetings + joint dataset workshop; majority agreement.
4	**Preprocessing**	T2W, TOF-MRA, and masks registered; standardized orientation; shared nnU-Net v2 plan.

Joint Workshop for Annotation Consensus

Distributed, Collaborative, & Federated learning workshop
In conjunction with MICCAI 2026





Distributed, Collaborative, & Federated learning workshop

Experimental Design

Institution-specific

- nnU-Net v2
- 1,000 epochs
- 3-fold cross-validation
- Evaluate on local fixed test set

Cross-site evaluation

- Transfer local model
- No fine-tuning
- No domain adaptation
- Evaluate on other hospitals

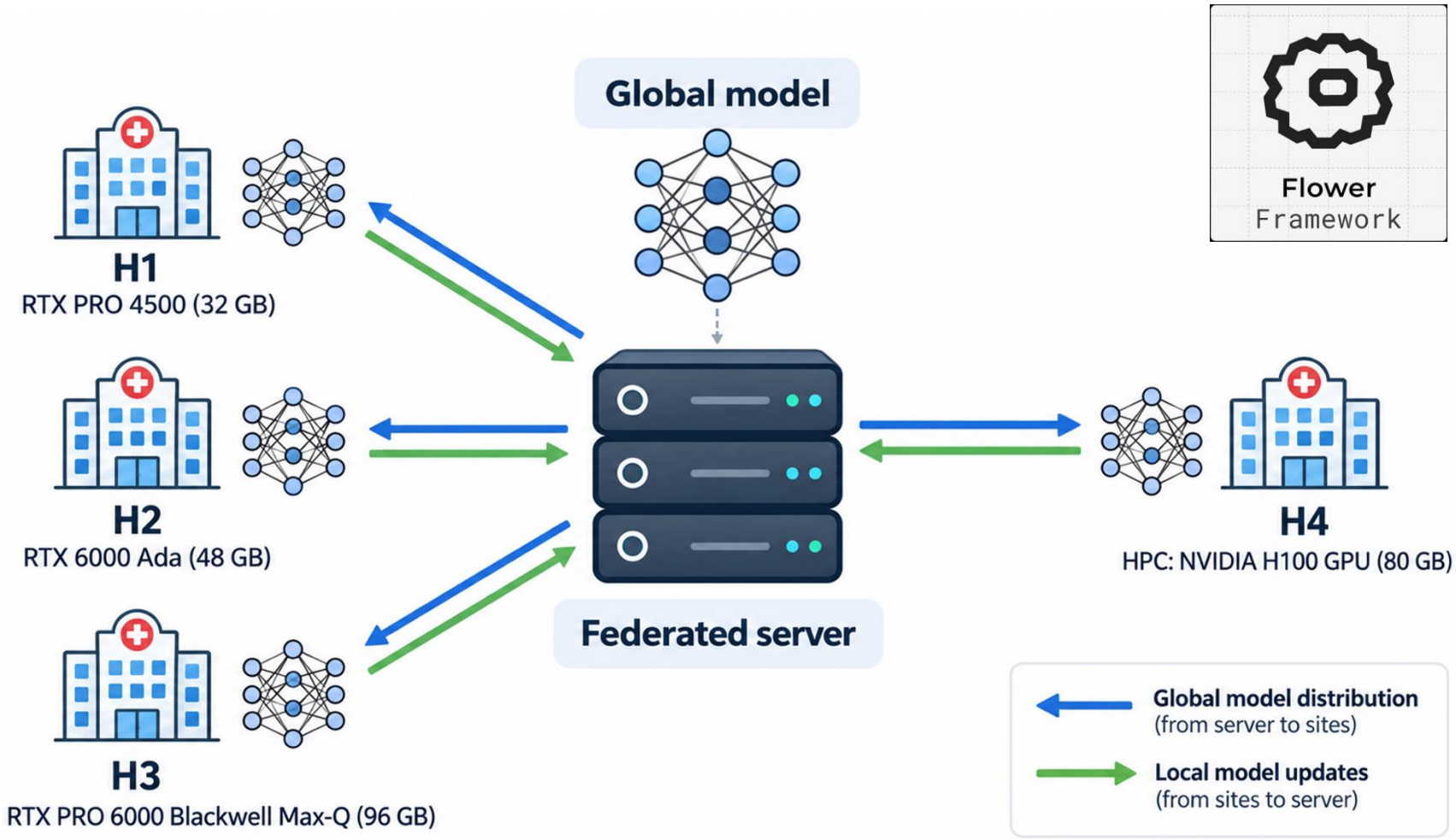
Federated training

- Flower framework
- Synchronous FedAvg
- 30 rounds × 10 local epochs
- All four hospitals each round

Controlled comparison The same inputs, preprocessing, architecture, loss, optimization settings, and fold assignments were used across local and federated experiments.

A pooled central benchmark was not feasible because approvals at three hospitals prohibited patient-level data pooling.

Federated Training in Real Hospital Infrastructure



- Only client model updates were transmitted to the coordinating server.
- Communication used TLS encryption.
- GPU resources differed substantially across hospitals.
- H1 connected through a commercial 5G modem because hospital-network use was not permitted.

RESULTS

Segmentation Performance Across Hospitals

Distributed, Collaborative, & Federated learning workshop

In conjunction with MICCAI 2026

Held-Out Test Sets

Training setting	H1	H2	H3	H4
H1 local	0.812	0.226	0.002	0.130
H2 local	0.802	0.546	0.039	0.281
H3 local	0.582	0.397	0.105	0.241
H4 local	0.754	0.399	0.056	0.308
Federated	**0.816**	**0.575**	**0.300**	**0.523**

Mean Dice on held-out site test cohorts

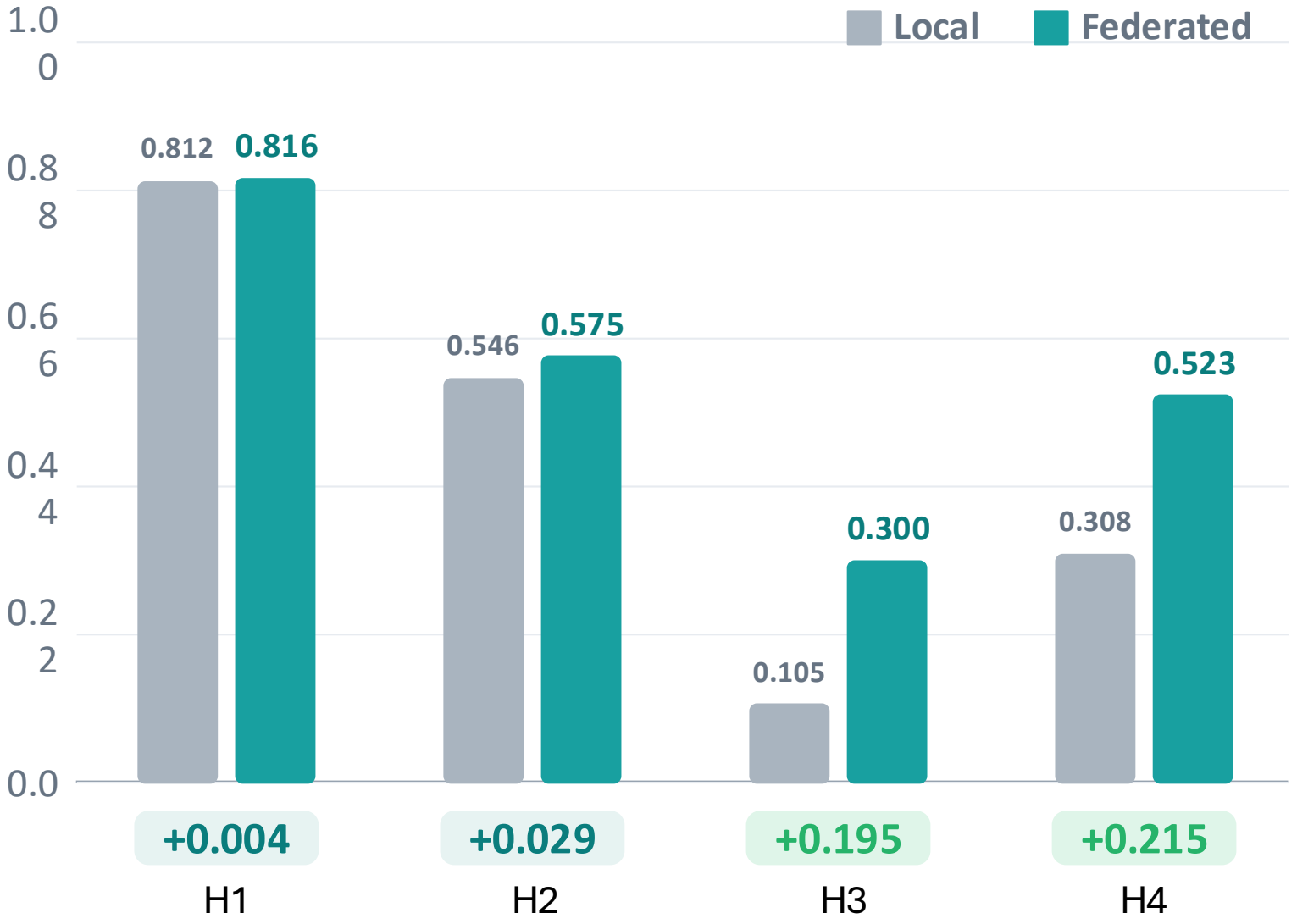
Cross-site generalization was limited and asymmetric

Example

H1 model: 0.812 on H1 → 0.002 on H3

RESULTS

Federated Training Improved Mean Dice at Every Site



MACRO-AVERAGED DICE

0.443 → 0.553

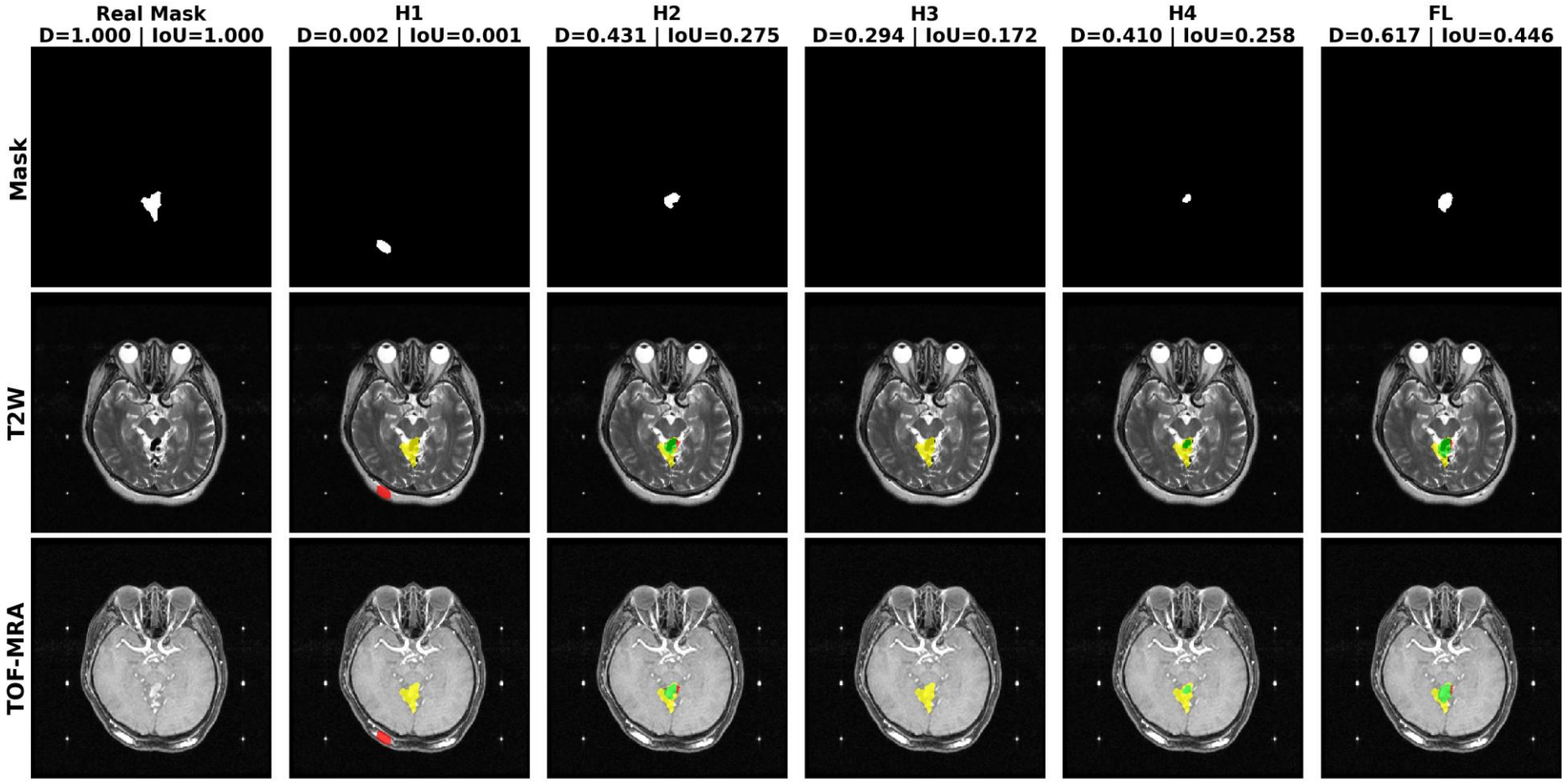
+0.110

95% bootstrap CI
0.050–0.175

Largest gains: H3 and H4
Inter-site variability remains.

RESULTS

Qualitative Comparison on a Held-Out H2 Case



Representative H2 held-out test case

Random H2 test case

Local H1	0.002
Local H2	0.431
Local H3 H3	0.294
Local H4 H4	0.410
Federated	**0.617**

Closer to the expert mask

DISCUSSION

Feasibility, Limitations, and Next Steps

Real-world feasibility

120 / 120
client updates
All scheduled updates received and aggregated.

≈ 62 h
training run
Completed using local site compute and networks.

0
patient files moved
Images, masks, and metadata stayed local.

Key limitations

- Only 10 held-out test cases per hospital.
- FedAvg was the only aggregation strategy evaluated.
- No independent external hospital for validation.
- No centrally pooled benchmark.
- No secure aggregation or differential privacy.
- Clinical impact metrics were not assessed.

Next steps

- Expand cases and external sites where feasible.
- Explore data-efficient strategies such as synthetic image augmentation.
- Evaluate clinically relevant outcomes and operational robustness.

Take-Home Message

- Single-site models do not generalize reliably across hospitals
- Federated learning improved segmentation at all four sites
- Real-world training was feasible without sharing patient data

Federated learning turns distributed rare-disease data into a shared advantage, improving multicenter bAVM segmentation while keeping data local.

Acknowledgements

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University of Illinois System

NSTC 國家科學及技術委員會 National Science and Technology Council

New collaborators are welcome

Project Page

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